Scalar curvature Rigidity

Recap

(M^n, g) Riem. mfd is spin iff $\omega_1(M) = 0$ and $\omega_2(M) = 0$

$\rightsquigarrow$ For every Cl_n-module W we can construct a $Cl(TM)$-module bundle $S \to M$ s.t.

$$S_x \overset{\text{isometric}}{\cong} W$$

$S \to M$ carries a compatible connection ∇^S induced by ∇^{LC}

$\rightsquigarrow$ $S \to M$ is a Dirac bundle in a canonical way

$E \to M$ Hermitian vector bundle with metric connection ∇^E

$\rightsquigarrow$ $S \otimes E \to M$ with $\nabla^{S\otimes E} := \nabla^S \otimes id + id \otimes \nabla^E$ is again a Dirac bundle

Schrödinger-Lichnerowicz formula

$$D_E^2 = \Delta + \frac{1}{4} scal_g + \mathcal{R}^E$$

$\mathbb{C}l_{2k} \cong Mat(2^k, \mathbb{C})$

$\mathbb{C}l_{2k+1} \cong Mat(2^k, \mathbb{C}) \oplus Mat(2^k, \mathbb{C}) \xrightarrow[p_{1/2}]{} Mat(2^k, \mathbb{C}) \Big\} \rightsquigarrow \mathbb{C}^{2^k} =: \Sigma_n$

$n = 2k$ or $2k+1$

$Cl_{2k} \hookrightarrow \mathbb{C}l_{2k} \longrightarrow Mat(2^k, \mathbb{C})$

Let $\Sigma M \to M$ be the spinor bundle associated with Σ_n

$\rightsquigarrow$ Assoc. Dirac op.

$$D : C^\infty(\Sigma M) \longrightarrow C^\infty(\Sigma M)$$

$E \to M$ Hermn. vector bundle with metric connection

$\rightsquigarrow D_E : C^\infty(\Sigma M \otimes E) \longrightarrow C^\infty(\Sigma M \otimes E)$

$\underline{n = 2k}$

Def $(e_1, \dots, e_n)$ oriented ONB of $\mathbb{R}^n$

$$\omega_{\mathbb{C}} := i^k e_1 \cdot \dots \cdot e_n \in \mathbb{C}l_n$$

is called the complex volume element

- $\omega_{\mathbb{C}}$ invariant under $SO(n)$

  $\rightsquigarrow \omega_{\mathbb{C}} \in C^\infty(\mathbb{C}L(TM))$

- $\omega_{\mathbb{C}}^2 = id_{\Sigma M}$
- $\nabla^S \omega_{\mathbb{C}} = 0$

$\Rightarrow$ orthog. splitting $\Sigma M \cong \Sigma M^+ \oplus \Sigma M^-$

($\Sigma M^\pm$: ± 1-eigenbundles of $\omega_{\mathbb{C}}$)

called chirality decomposition of ΣM

$\rightsquigarrow D_E^{\pm} : C^\infty(\Sigma M^{\pm} \otimes E) \longrightarrow C^\infty(\Sigma M^{\mp} \otimes E)$

$D = \sum_i e_i \cdot \nabla^S_{e_i} \qquad D\omega_{\mathbb{C}} = -\omega_{\mathbb{C}} D$

$$D_E = \begin{pmatrix} 0 & D_E^- \\ D_E^+ & 0 \end{pmatrix}$$

M closed (compact without boundary), then

- D_E extends to a self-adj. op.

$$\bar{D}_E : H^1(\Sigma M \otimes E) \longrightarrow L^2(\Sigma M \otimes E)$$

- $\bar{D}_E$ is Fredholm

$\rightsquigarrow \operatorname{index}(\bar{D}_E) = \dim(\ker(\bar{D}_E)) - \underbrace{\dim(\operatorname{coker}(\bar{D}_E))}_{=\dim(\ker(\bar{D}_E^*))} = 0$

$\rightsquigarrow$ take instead

$\operatorname{index}(\bar{D}_E^+) = \dim(\ker(\bar{D}_E^+)) - \dim(\ker(\bar{D}_E^-))$

Thm (Atiyah-Singer)

$$\operatorname{index}(\bar{D}_E^+) = \int_M \hat{A}(TM) \wedge \operatorname{ch}(E) = \langle \hat{A}(TM) \cup \operatorname{ch}(E), [M] \rangle$$

$\hat{A}(TM) \in \Omega^{4*}(M)$, $\operatorname{ch}(E) \in \Omega^{2*}(M, \mathbb{C})$

Scalar curvature

Def: (M^n, g) Riem. mfd., $p \in M$

$$\frac{\mathrm{Vol}(B_\varepsilon(p) \subset M)}{\mathrm{vol}(B_\varepsilon(0) \subset \mathbb{R}^n)} = 1 - \frac{1}{6(n+2)} \mathrm{scal}_g(p)\,\varepsilon^2 + O(\varepsilon^3) \qquad \varepsilon \searrow 0$$

Example:

	$\mathbb{R}^n$ (g_{Eucl})	S^n (g_0)	$\mathbb{H}^n$ ($g_{\mathbb{H}}$)
scal	0	$n(n-1)$	$-n(n-1)$

Thm (Kazdan-Warner '75)

M closed mfd. $\dim(M) = n \geq 3$, $f \in C^\infty(M)$ s.t. $\exists x \in M : f(x) < 0$

$\Rightarrow \exists$ Riem metric g with $\mathrm{scal}_g = f$

Thm (Lichnerowicz)

M closed spin mfd. s.t. $\hat{A}(M) = \int_M \hat{A}(TM) \neq 0$

$\Rightarrow M$ is not psc

Proof Let $u \in \ker(D)$ $\qquad$ $\mathrm{index}(D) = \hat{A}(M)$

$$(u,v)_{L^2} = \int_M \langle u, v \rangle_x \, d\mathrm{vol}(x)$$

$$\Rightarrow 0 = (D^2 u, u)_{L^2}$$

$$= \left(\Delta u + \tfrac{1}{4}\mathrm{scal}_g u, u\right)_{L^2} \qquad (\Delta = \nabla^*\nabla)$$

$$= \underbrace{\|\nabla^S u\|^2_{L^2}}_{\geq 0} + \tfrac{1}{4}(\mathrm{scal}_g u, u)_{L^2} > 0 \quad \lightning$$

If $\mathrm{scal}_g > 0 \Rightarrow \lightning$ $\square$

Thm 1 (Llarull '98)

(M^n, g) closed connected spin n-fold, $n \geq 2$

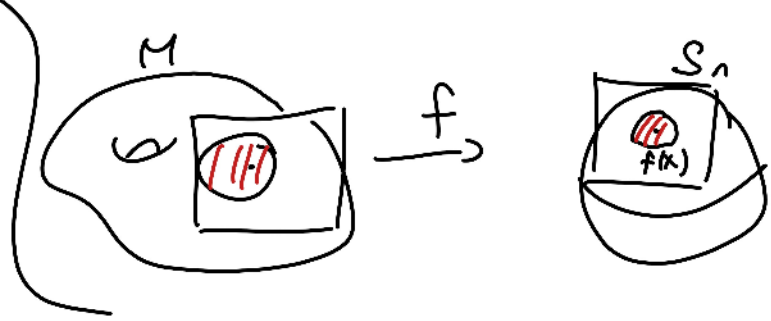
a) $\mathrm{scal}_g \geq n(n-1) = \mathrm{scal}_{S^n}$

b) $f : (M, g) \to (S^n, g_0)$ smooth, $\deg(f) \neq 0$ s.t.

- f 1-Lipschitz $\quad n = 2 \quad$ — $|df| \leq 1$
- f area-nonincreasing $\quad n \geq 3$

  ↳ $\Lambda^2 df : \Lambda^2 TM \to \Lambda^2 TS^n$, $|\Lambda^2 df| \leq 1$

$\Rightarrow$ f Riem isometry



special case $M = S^n$, $f = id : (S^n, g) \to (S^n, g_0)$

$\rightsquigarrow g = g_0$

Proof of thm 1 /2

$(\Sigma M, \nabla) \to M$

$(\Sigma S^n, \nabla^0) \to S^n$

} spinor bundles

w.r.t. smooth background metric

$(E, \nabla^E) := f^* (\Sigma S^n, \nabla^0)$ Lipschitz bundle

$\to D_E : \overset{\text{Lip}}{C^\infty}(\Sigma M \otimes E) \to \overset{L^2}{C^\infty}(\Sigma M \otimes E)$

1) Atiyah-Singer

(*) approx. g by smooth metrics → cont. family of of Dirac operators

n even $\quad \Sigma M \equiv \Sigma M^+ \oplus \Sigma M^-$

$\overline{D}_E$ is Fredholm

$$\operatorname{index}(\overline{D}_E^+) \overset{(*)}{=} \langle \hat{A}(TM) \cup ch(E), [M] \rangle$$

$$= \ldots = \underbrace{\deg(f)}_{\neq 0} \cdot \underbrace{\chi(S^n)}_{=2} \neq 0$$

$\Rightarrow \exists$ harmonic spinor $u \in \ker(D_E)$

n odd: consider Dirac op. on $(M \times S^1, g + r^2 g_{S^1})$

$\rightsquigarrow$ harm. spinor of D_E on M

$r \to \infty$

2) Schrödinger-Lichnerowicz

$u \in \ker(D_E) \not\subset C^\infty(\Sigma M \otimes E)$

$$0 = \|D_E u\|^2_{L^2} \overset{\hat{H}}{=} \left((\overline{D}_E^2 u, u)_{L^2} \right)$$

distr. sense ↓

$$= \left(\underbrace{(\nabla^* \nabla u, u)_{L^2}}_{= \|\nabla u\|^2_{L^2} \geq 0} \right) + \underbrace{\frac{1}{4}(\operatorname{scal}_g u, u)}_{\geq \frac{1}{4} n(n-1) \|u\|^2_{L^2}} + \underbrace{(\mathcal{R}^E u, u)_{L^2}}$$

≥ 0

At a.e. every $x \in M$

$$\langle \mathcal{R}^E u, u \rangle = \sum_{i<j} \langle e_i \cdot e_j \cdot R^E(e_i, e_j) u, u \rangle$$

$$= \sum_{i<j} \langle e_i \cdot e_j \, (f^* R^{S^n})(e_i, e_j) u, u \rangle$$

$$= \sum_{i<j} \langle e_i \cdot e_j \otimes f_*(e_i) \cdot f_*(e_j) \cdot u, u \rangle$$

Sing. value decomp.:

$\exists (\varepsilon_1, \dots, \varepsilon_n)$ ONB of $T_{f(x)} S^n$ and $\mu_1, \dots, \mu_n \in \mathbb{R}^n$ s.t.

$$f_*(e_i) = \mu_i \varepsilon_i$$

f 1-Lipschitz $\Rightarrow$ $0 \le \mu_i \le 1$ only if $n = 2$

$$= \sum_{i<j} \mu_i \mu_j \langle e_i e_j \otimes \varepsilon_i \cdot \varepsilon_j \, u, u \rangle$$

$$\geq - \sum_{i<j} \underbrace{\mu_i \mu_j}_{\le 1} |u|^2$$

$$\geq - \frac{1}{4} n(n-1) |u|^2$$

$\Rightarrow \operatorname{Scal}_g \equiv n(n-1)$, $\mu_i \mu_j = 1 \quad \forall i \neq j$

$\Rightarrow \mu_i = 1 \quad \forall i$

$\Rightarrow f$ local isometry a.e.

($\Rightarrow f : (M, g) \to (S^n, g_0)$ Riem. covering

$\Rightarrow f$ Riem. isometry)

□

df orient. preserv. a.e.

$\Rightarrow$ f metric isometry

Reshetnyak ☐

Thm 2 (Cecchini-Hanke-Schick '22, C-H-S-S '25)

M^n smooth closed connected spin mfd., $n \geq 2$ even, odd

a) $g \in W^{1,p}(T^*M \otimes T^*M)$, $p > n$ admissble Riem. metric s.t. $\mathrm{scal}_g \overset{\text{distr.}}{\geq} n(n-1)$

b) $f : (M, g) \longrightarrow (S^n, g_0)$ Lipschitz, $\deg(f) \neq 0$ s.t.

- f 1-Lipschitz $n = 2$
- f area-nonincreasing a.e. $n \geq 3$

$\Rightarrow$ $f : (M, d_g) \longrightarrow (S^n, d_0)$ metric isometry